

BREATHTURES: A First Step Towards Breathing Gestures as Distinct Input Modality

Lisa Burr

lisaanneke.burr@plus.ac.at

Department of Artificial Intelligence and Human
Interfaces, University of Salzburg
Salzburg, Austria

Julia Mayrhauser

julia.mayrhauser@stud.sbg.ac.at

Department of Artificial Intelligence and Human
Interfaces, University of Salzburg
Salzburg, Austria

Julius Šula

julius.sula@plus.ac.at

Department of Artificial Intelligence and Human
Interfaces, University of Salzburg
Salzburg, Austria

Alexander Meschtscherjakov

alexander.meschtscherjakov@plus.ac.at

Department of Artificial Intelligence and Human
Interfaces, University of Salzburg
Salzburg, Austria

ABSTRACT

The use of breath as interaction modality has sparked some interest recently in HCI. Research in this field actively explored breathing interaction (e.g., guidance and feedback) in different contexts. So far, very little systems can be found that systematically use defined breathing gestures as distinct input modality. In this work, we describe the design, development and implementation of five distinct breathing gestures (BREATHTURES), that can be applied as defined input modality to interact with an application. We present a qualitative user study (n=7) focusing on usability and user experience of the developed BREATHTURES when applied in a game-like application. Results indicate an overall positive rating in terms of fun and usability. Some BREATHTURES were perceived harder to perform and to detect than others. Further iterations of our work will require refined BREATHTURE detection. Overall, our work demonstrates the design, implementation and application of pre-defined breathing gestures as distinct input modality.

CCS CONCEPTS

• **Human-centered computing** → **Gestural input**; *User interface design*; Usability testing.

KEYWORDS

interaction, breath detection, breathing pattern, distinct input modality, modular environment

ACM Reference Format:

Lisa Burr, Julius Šula, Julia Mayrhauser, and Alexander Meschtscherjakov. 2023. BREATHTURES: A First Step Towards Breathing Gestures as Distinct Input Modality. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3544549.3585787>

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '23, April 23–28, 2023, Hamburg, Germany

© 2023 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9422-2/23/04.

<https://doi.org/10.1145/3544549.3585787>

1 INTRODUCTION

Breathing, as a basic vital function comes naturally and unconscious to most of us. Out of our physiological functions' respiration is one that we can also consciously control [36]. Our breath holds many different patterns and distinguishes itself in various qualities such as amplitude or frequency [12]. Hence, we can intentionally use our breath and its variety to naturally interact with the world around us.

This understanding has sparked some interest in using respiration as interaction modality in HCI. Different technologies and sensors have been designed and researched in order to access breathing data and allow us to obtain specific respiration parameters (e.g., breathing rate and depth, piezoelectric signal, air temperature). Breath interaction systems, that make use of these technologies and sensors, evolved in different contexts and settings, e. g. in rehabilitation [10, 32], to support people with disabilities [35], enhance the way we interact with systems nowadays [31], or deepening the immersion within the usage of a system or game [23, 29, 30]. Breathing as a direct control mechanism has recurrently been applied as input modality to actively influence elements, situations, or events in games or game-like experiences, or specific user interfaces. However, only few examples can be found that intentionally apply distinct breathing techniques or patterns as direct input controlling mechanism [13, 29, 30].

This motivates our work in focusing on the design and development of distinct breathing gestures and their direct application in one specific context. To access respiratory data, we used a sensor shirt that allowed us to access thoracic, as well as abdominal movement. Thus, we had the chance to utilize the full scope of breath inducted movements in those two areas for our definition and detection of breathing gestures, henceforth referred to as BREATHTURES.

Within our work, we explored natural breathing patterns and techniques and developed five different BREATHTURES to be utilized as a distinct input modality. We then designed a breath-controlled game-like application for Android where every BREATHTURE was implemented and mapped to a specific action. The application was designed as interactive experience for the user to try out different breath inputs.

The contribution of this work is thus twofold:

- We designed and developed an initial set of five distinct BREATHTURES.
- We implemented the BREATHTURES within a game-like Android application and present the results of a user study investigating first insights regarding usability and user experience.

2 RELATED WORK

Different methods have been described for accessing and using breath as an input method in HCI, including using a microphone to detect nasal exhalation input [14, 19, 20] or blowing actions [2, 7, 16, 22], using high frequency wireless signals [1], to obtain physiological signals or using a worn (stretch) sensor that detects chest and/or abdominal movement [5, 33].

Continuous research in this field has led to a growing number of games, systems and applications that apply respiration either as indirect or direct interaction modality. Breath as a indirect control mechanism is mostly found in applications focusing on mindfulness. These systems usually aim at inducing “beneficial” breathing patterns or enhance experience in users [15, 21]. Breath as a direct control mechanism is instead applied to use breathing input to actively influence elements, situations, or events to operate e.g. desktop actions like scrolling or dragging [22, 26], or to manipulate mobile phone [17] and smart watch [6] user interfaces. Also, games and game-like experiences have gradually implemented breath data as additional direct input modality to manipulate game elements, enhance immersion and engagement for players, or support rehabilitation processes [29, 30, 32]. Distinctions between different types of breathing patterns have been used to identify respiratory diseases [34] or to recognize stress [8].

In HCI, however, the selective use of distinct breathing patterns or techniques as a direct input modality has been investigated relatively less. Work in this context has primarily been focused on virtual reality games. In the design of their virtual reality action game “Bubble Tower” Heng and Albert [13] differentiate between exhaling patterns in terms of speed and intensity of the breath. Their breathing input design consists of breath-in, hold, lasting breath-out, blast-out and break. They describe a preliminary user test and by its outcome conclude that the consistency of the breathing sensor needs further improvement to achieve a more accurate reading. Sra et al. [29] describe the design of four breathing actions (a forceful exhalation (gust), a sustained strong exhalation (gale), slowly blowing out air through an open mouth (waft) and no exhalation (calm)) each different in magnitude and length, that are mapped to actions and effects in two different games. The authors conclude, that breathing actions can create exciting and immersive game-play experiences. For them breathing has three essential “forms”, i. e., inhalation, exhalation and hold, however they did not cover inhalation in their breathing actions. Tatzgern et al. [30] adopt the same breathing actions to demonstrate that their developed breathing engine, that translates sensor measurements into forces that are applied to virtual objects in real-time, can robustly differentiate between breathing patterns. Both research groups outline design

guidelines for using breathing actions and creating experiences in game-play.

With our focus on developing a set of pre-defined breathing gestures, that enable the use of breath as distinct input modality beyond the gaming context, we step away from virtual reality scenarios. In addition to the natural breathing “forms” found in [29] and trades described by [9], existing design recommendations [29, 30] are taken into account in our BREATHTURE development process.

3 BREATHTURES

Human breathing inhibits complex and highly individual patterns which distinguish themselves in various characteristics like amplitude and frequency, but also regarding the involved areas of the body e. g., abdomen (i.e. stomach) or thorax (i.e. chest) [12]. To allow the discrimination between abdominal and thoracic breathing, we chose to gather breathing data via a sensor shirt with respiration sensors that cover both respective regions (see section 4.3.1).

3.1 Definition

Building on the knowledge that humans can directly and consciously manipulate their own breath characteristics, we decided to start the BREATHTURE development process by conducting a collaborative brainstorming session. This session was carried out by an interdisciplinary team of human factors engineers, sport scientists, interaction designers, and computer scientists, aiming to gather a number of natural and intuitive breathing actions (e. g., smoking, yawning, coughing). Next, we fragmented the collected breathing actions into smaller traits to extract measurable properties. Inspired by the outlined breathing traits in [9], we decided to focus on measuring breathing length, depth, pace, holding time and (body)location. These traits were further on used as baseline information to define possible BREATHTURES.

Considering design recommendations for natural breathing interaction in the context of virtual reality [29, 30] we defined a set of initial ideas. By testing their feasibility within the group, the set was limited down to five final BREATHTURES: (1) *hold breath*, (2) *deep abdominal breath*, (3) *deep thoracic breath*, (4) *abdominal staccato breath*, and (5) *thoracic sigh*.

3.2 Design

Every BREATHTURE is defined through its respective amplitude and frequency. Additionally, abdominal and thoracic breathing is distinguished. The respective thresholds for reaching a specific amplitude and frequency were defined relatively to the users individual calibration (see section 3.3.1) values. The different breathing gestures were designed as follows:

3.2.1 Hold breath. Breath holding is a natural breathing trait we detected by monitoring expansion and deflation values of the abdominal or thoracic region. Hold breath was detected, when the difference between the currently measured values and the previous values were below the calibration threshold, for a defined amount of time. To improve recognition of breath holding and prevent influences by outliers in the data, a 20% tolerance buffer was added. The calibrated threshold was defined by measuring the breathing values for five seconds while inhaled or exhaled respectively and measuring the maximal delta between two consecutive breathing

values. When the delta of consecutive breathing readings were below this calibrated value for two seconds a hold BREATHTURE was detected.

3.2.2 Deep abdominal/thoracic breath. We distinguished between a deep abdominal and a deep thoracic breath. The individual "deep breath" was defined in the calibration phase, where every user was asked to take a deep breath using primarily their abdomen or primarily their thorax, versus a regular breath. Values captured in the calibration process defined the threshold which was further on used for the detection of the deep abdominal and thoracic BREATHTURES. To obtain the calibrated threshold we measured the local maxima of the breathing value during the calibration of a single deep breath through the thorax and abdomen respectively and calculated the median of them.

3.2.3 Abdominal staccato breath. Staccato is a musical term and translates from Italian to "detached". It describes a tone that is played shortly and is often followed by silence. This gave the inspiration for this BREATHTURE, where two short and rapid inhales through the abdomen are executed and detected. To detect "rapid" breath changes we implemented a buffer of four values. If the last value is 45% bigger than the first, a rapid inhale is detected. This accounts only when the user is reaching at least 35% of their maximal breathing depth.

3.2.4 Thoracic sigh. A sigh can have many forms with differences in frequency and amplitude. In this work, a thoracic sigh was detected when a deep thoracic inhale was followed by a rapid thoracic exhale. Detecting the sigh works similar to the staccato detection. Using a 10 value buffer, the first (bigger) and last (smaller) value have to show a difference of at least 75%.

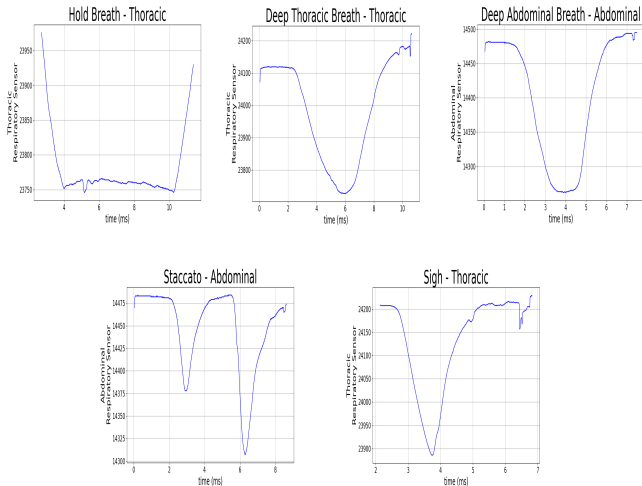


Figure 1: Visualization of breathing sensor data for all five BREATHTURES. Top left: hold breath; top middle: deep thoracic breath, top right: deep abdominal breath; bottom left: abdominal staccato breath; bottom right: thoracic sigh.

3.3 Implementation

To apply the designed BREATHTURES in a specific context, we implemented them in a breath-controlled game-like application for Android. Coinciding with the work of Tatzgern et al. [30] we avoided extensive tutorials, by introducing the different BREATHTURES and linked actions practically within the calibration process. In line with the design strategies of Sra et al. [29] we aimed to design all actions to feel natural and relevant to the BREATHTURE that triggered them.

3.3.1 Calibration. Main purpose of the calibration was to record the individual minimal and maximal breathing values of every user. Therefore, the user was visually instructed to take (1) five regular, (2) five deep abdominal, (3) five deep thoracic and (4) five regular breaths. The local minima and maxima of each breath were used to calculate the median for abdominal and the thoracic values. These were further on set to define the individual maximal and minimal thresholds in the subsequent BREATHTURE detection. After threshold definition, every BREATHTURE was explained to the user and thereupon practically executed.

3.3.2 Actions. We implemented five different actions with accompanying animations in our game-like application: (1) select menu option, (2) blowing snow away, (3) blowing clouds away, (4) let birds fly, and (5) start rocket. They were all mapped to one distinct BREATHTURE (see figure 2).

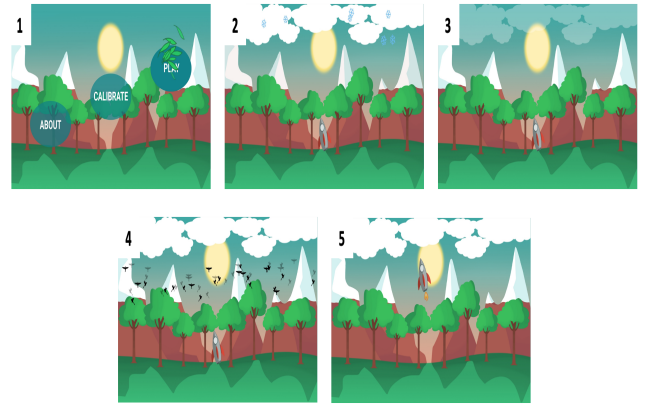


Figure 2: Overview of actions and accompanying animation that are mapped to the five developed BREATHTURES. (1) Hold breath to select menu option; (2) deep abdominal breath to blow falling snow away; (3) deep thoracic breath to blow clouds away; (4) abdominal staccato breath to let birds fly out of the trees; (5) thoracic sigh to let rocket start into the air.

3.3.3 Modular Environment. To take a first step towards a *breathing gesture set* that is functioning with different sensors and in different application contexts, the technical implementation was thoughtfully designed to allow a modular environment. Meaning that all components that directly rely on the sensor data are only loosely coupled. This was achieved by structuring the architecture into four main components: (1) Data Connection, (2) Calibration,

(3) Gesture Detection and (4) Visual Interface. The Sensor as fifth component is independent from the system.

4 METHODS

To get a first insight into the usability and user experience of the developed BREATHTURES, they were implemented in game-like Android application. Thereupon, everything was tested in a preliminary user study¹ (n=7). All of tests were conducted on-site at the department of human-computer interaction of the University of Salzburg to guarantee a standardized environment.

4.1 Participants

Seven participants were recruited via mouth-to-mouth and the mail distributor of the University of Salzburg and University of Applied Science Salzburg. Six participants were between the ages of 20 to 30 years and one participant in between 50 and 60 years. All participants met the recruitment criteria of a age range from 16-65 years and did not report any of the following exclusion criteria (systemic diseases or other known diseases of the organs, diabetes II, neurological or psychological diseases of any kind, current medical or psycho-therapeutic treatment, alcohol or drug abuse, allergies - especially contact allergies to plasters, creams etc. or pollen allergy, Covid-19 infection within the last three months, long Covid-19 symptoms). All participants were tested following the same study protocol. The duration of the whole experiment was approximately 30 minutes.

4.2 Study Procedure

Participants were welcomed in the lab and, after clarifying open issues, asked to read and sign an informed consent form, stating the general purpose of the research, listing all of the above listed exclusion criteria and their willingness to participate in this study. In order to be able to assess previous experience with wearable and smartphone controls a brief interview was conducted at the beginning. Following, participants were introduced to the study procedure and equipped with the Hexoskin sensor shirt (see section 4.3.1). Participants were asked to stand upright, if possible, throughout the whole study. The test scenario started with the calibration process (see section 3.3.1), where individual breath thresholds were defined and all five BREATHTURES were explained and demonstrated in detail. Following the calibration, participants were instructed to use their breath to navigate throughout the menu and start the game to explore the application. They were told to freely try out all the different BREATHTURES to discover their assigned actions. In addition to this, participants were asked to think out loud during the interaction. It was moreover stated that they could ask for help at any time if needed. All observations, comments and thoughts were noted down by the researchers. The interaction ended when the participant selected the "END" item on the respective screen. The preliminary user study was concluded by completing a set of questions assessing usability and user experience aspects of the calibration process and BREATHTURE interaction. Perceived ease of executing the calibration and different BREATHTURES was assessed via Single Ease Question (SEQ) [28] on a scale from difficult

(1) to easy (5) ("Overall, how difficult or easy was it for you to execute the following BREATHTURE?"). To compare and exploit the effectiveness of the different BREATHTURES, participants were asked to point out which worked particularly well for them, which did not work well and which took more than one try to be executed. Here, participants were able to mention multiple options. Lastly participants were invited to rate, on a scale from no fun (1) to a lot of fun (10), how much fun they had interacting with the system.

4.3 Experimental Setup

4.3.1 Breath Sensor. To access respiration data via Bluetooth, participants were equipped with a Hexoskin sensor shirt (Carre Technologies, Canada), with sensors for activity (3-axis accelerometer, ± 16 g, 64 Hz), respiration (dual 2-channel respiratory inductance plethysmography (RIP), 16 bit, 128 Hz), and heart (1-lead ECG, 12 bit, 256 Hz). The shirt has two stretch sensors, one in the abdominal and one in the thoracic region.

4.3.2 Android Application. To enable an interactive breath interaction, the Hexoskin SDK for Android was utilized to develop the application. The application was run on a Samsung A52s 5G. During the study, the phone was placed at eye level of the participant.

5 RESULTS AND DISCUSSION

The aim of our study was to gather first usability and user experience insights regarding the application of the five BREATHTURES as distinct input modality. Following, we present and discuss our key findings separately for the calibration process and the distinct use of the five BREATHTURES in the game-like context.

5.1 Calibration

Overall, participants stated that following the instructions during the calibration process was easy to them (mean=3.85; SD=1.21). However, some also indicated, that executing the given breathing tasks in the right way might take some practice for them. Here, especially the differentiation between thoracic and abdominal breathing was mentioned. This feedback is consistent with our observations, pointing out that most of the participants had trouble breathing predominantly into one of both locations. Highlighting the importance of taking into account individuals natural breathing patterns [4] we believe that breathing gestures should be adjustable to the user. This goes in line with suggestions of [29] to customize breathing actions for different abilities. We further on suggest that breathing awareness and skills for specific breathing patterns should be accessed and, if necessary, trained properly before they are utilized as natural input modality. As perceptually extending breath awareness is already embraced within a large body of systems that utilize breath data [25], existing breathing training and/or mindfulness applications could be leveraged to establish an individual foundation for utilizing breath as input modality.

5.2 BREATHTURES

SEQ: *Hold breath* with a mean of 4.57 (SD=0.79) and *thoracic sigh* with a mean of 4.29 (SD=0.49) were rated the easiest by the participants. However, also *deep abdominal* (mean=3.29; SD=1.25) and *thoracic* (mean=3.29; SD=1.5) and *abdominal staccato breath* (mean=3.14;

¹For this research ethical approval was granted by the Ethics Committee of Salzburg University, reference number: GZ 12/2022

SD=1.87) ranked over average.

Which BREATHTURE worked well?: *Hold breath* and *deep thoracic breath* have worked well for five out of seven participants. Whereas *abdominal staccato breath* worked well for four, *deep abdominal breath* for three and the *thoracic sigh* only for two participants (see figure 3).

Which BREATHTURE did not work well?: When asked this question, five participants choose *deep abdominal breath*. This fact got also reported by participants thinking aloud during the interaction. Some further on stated that it did not feel possible for them to deeply breath primarily into their abdomen. This matches our previously mentioned observations during the calibration. However, as deep and slow abdominal breathing is seen as an important skill for the management of stress and also as strategy in order to work with pain [27], we see a high value in constantly practicing deep breathing (at a pleasant level) in context of using the respective BREATHTURE as distinct input. *Abdominal staccato breath* and *thoracic sigh* both did not work well for two out of seven participants. One participant rated *deep thoracic breath* as not working well. Interestingly, no participant mentioned that *hold breath* did not work well.

Did any of the BREATHTURES take more than one try to be detected?: *Deep abdominal* and *thoracic*, as well as *abdominal staccato breath* took more than one try for four participants. *Thoracic sigh* was mentioned by three. The detection of *hold breath* seemed to have worked quite well, with only one participant stating that more than one try was needed.

How much fun did you have interacting with this system?: A mean value of 9 (SD=1.15) points out, that the system was really fun to use for the participants, even though minor setbacks with ambiguity and detection of the BREATHTURES have been reported.

Looking at the different scores in the ratings of BREATHTURES working well or not well, we assume that it can be difficult to rate the pure success of breath interaction. Thus, specific criteria should be specified that define the success of using a BREATHTURE in a certain use case. In this context we also want to address the effect that our body position can have on the way we breathe [3] and thus on our ability or success to execute certain BREATHTURES. The importance of body posture has already been highlighted in the scope of breathing exercise game design [23]. We recommend to take this knowledge in account when designing for breath as distinct input modality. The number of BREATHTURES that took more than one try to be detected (see figure 3) indicate a clear need for a refined BREATHTURE detection. Besides considering the quality and placement of the sensor as previously concluded by different research teams [13, 23, 30], we are aiming to apply enhanced algorithms for a more personalized detection and calibration process in future work. Overall, the generally positive results are supportive to the use of breathing as distinct input modality, to trigger certain actions within an application.

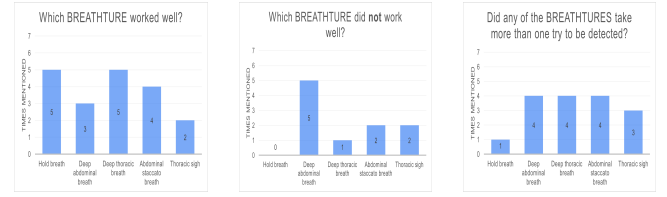


Figure 3: Visualisation of preliminary study results. Left: "Which BREATHTURE worked well?" Middle: "Which BREATHTURE did not work well?" Right: "Did any of the BREATHTURES take more than one try to be detected?"

6 LIMITATIONS AND FUTURE WORK

This work presents five distinct breathing gestures functioning as distinct input modality to interact with an application. Findings regarding their implementation are so far limited to one preliminary usability evaluation with seven participants in one use-context (game-like Android application). Further studies should extend the scope of this work by expanded and more comprehensive evaluation. However, we acknowledge that for usability purposes seven participants can already give an indicative first insights [18] for upcoming iteration cycles. There are several limitations in our current gesture detection that could be overcome. Detectable differences between BREATHTURES have turned out to be rather blurry, e. g., when people exhale after a deep breath, they often do it in a similar fashion as sighing. In context of future work, we consider the use of refined algorithms for a more personalized BREATHTURE detection and calibration process. The technical implementation was designed to allow a modular and expandable environment. Our next step to extend the scope of this research is to add further breathing gestures to the existing environment. Moreover, we would like to explore the use of breathing gestures as input modality while being in motion. The Hexoskin breath sensor shirt has explicitly been evaluated for valid breath measurement in motion e. g., running [11, 24]. Thus, we can envision future applications that incorporate our BREATHTURES in the context of sport and rehabilitation.

7 CONCLUSION

In this work, we described the design and implementation of five distinct breathing gestures (BREATHTURES). We demonstrated their application within an game-like Android app, specifically developed for this purpose. A preliminary evaluation revealed overall positive results in terms of usability and user experience of the interaction. However, we realised that further refinement of our breathing gesture detection is needed to ensure a successful interaction experience. Considering our findings, we have opened up a variety of directions for future work and envisioned applications.

ACKNOWLEDGMENTS

This work was partly funded by the Austrian Federal Ministry for Climate Action, Environment, Energy, Mobility, Innovation and Technology, the Austrian Federal Ministry for Digital and Economic Affairs, and the federal state of Salzburg.

REFERENCES

- [1] Fadel Adib, Zachary Kabelac, Hongzi Mao, Dina Katabi, and Robert C. Miller. 2014. Demo: Real-Time Breath Monitoring Using Wireless Signals. In *Proceedings of the 20th Annual International Conference on Mobile Computing and Networking* (Maui, Hawaii, USA) (*MobiCom '14*). Association for Computing Machinery, New York, NY, USA, 261–262. <https://doi.org/10.1145/2639108.2641756>
- [2] Vasundhara Agrawal, Varnika Naik, Mayuri Duggirala, and Sandeep Athavale. 2020. Calm a mobile based deep breathing game with biofeedback. In *Extended Abstracts of the 2020 Annual Symposium on Computer-Human Interaction in Play*, 153–157.
- [3] Ahmet Baydur, Rodney H Adkins, and Joseph Milic-Emili. 2001. Lung mechanics in individuals with spinal cord injury: effects of injury level and posture. *Journal of Applied Physiology* 90, 2 (2001), 405–411.
- [4] Gila Benchetrit. 2000. Breathing pattern in humans: diversity and individuality. *Respiration physiology* 122, 2-3 (2000), 123–129.
- [5] Lisa Burr, Nick Betzlbacher, Alexander Meschtscherjakov, and Manfred Tscheligi. 2022. Breathing Training on the Run: Exploring Users Perception on a Gamified Breathing Training Application During Treadmill Running. In *Persuasive Technology: 17th International Conference, PERSUASIVE 2022, Virtual Event, March 29–31, 2022, Proceedings* (Doha, Qatar). Springer-Verlag, Berlin, Heidelberg, 58–74. https://doi.org/10.1007/978-3-030-98438-0_5
- [6] Wei-Hung Chen. 2015. Blowatch: Blowable and hands-free interaction for smartwatches. In *Proceedings of the 33rd Annual ACM Conference Extended Abstracts on human factors in computing systems*, 103–108.
- [7] Yeqing Chen, Yulong Bian, Wei Gai, and Chenglei Yang. 2020. Adaptive Blowing Interaction Method Based on a Siamese Network. *IEEE Access* 8 (2020), 115486–115500.
- [8] Youngjun Cho, Nadia Bianchi-Berthouze, and Simon J Julier. 2017. DeepBreath: Deep learning of breathing patterns for automatic stress recognition using low-cost thermal imaging in unconstrained settings. In *2017 seventh international conference on affective computing and intelligent interaction (acii)*. IEEE, 456–463.
- [9] Jérémy Frey, May Grabli, Ronit Slyper, and Jessica R Cauchard. 2018. Breeze: Sharing biofeedback through wearable technologies. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*, 1–12.
- [10] N Garcia-Hernandez, I Muñoz-Pepi, F Villafranca-Martinez, and V Parra-Vega. 2022. Usability and Effectiveness Evaluation of a Game-Based Biofeedback System to Strengthen Respiratory Muscles. *International Journal of Human-Computer Interaction* (2022), 1–12.
- [11] Eric Harbour, Michael Lasshofer, Matteo Genitrini, and Hermann Schwameder. 2021. Enhanced breathing pattern detection during running using wearable sensors. *Sensors* 21, 16 (2021), 5606.
- [12] Eric Harbour, Thomas Stöggel, Hermann Schwameder, and Thomas Finkenzeller. 2022. Breath Tools: A Synthesis of Evidence-Based Breathing Strategies to Enhance Human Running. *Frontiers in Physiology* (2022), 233.
- [13] Oo Jin Heng and Quek Albert. 2020. Bubble Tower: Breathing Based Virtual Reality Action Game. In *Proceedings of the 2020 the 4th International Conference on Big Data and Internet of Things*, 33–37.
- [14] Rohan Hundia and Aaron Quigley. 2019. BreathIn: A breath pattern sensing approach for user computer interaction. In *Proceedings of the 31st Australian Conference on Human-Computer-Interaction*, 581–584.
- [15] Annkatrin Jung, Miquel Alfaras, Pavel Karpashevich, William Primett, and Kristina Höök. 2021. Exploring Awareness of Breathing through Deep Touch Pressure. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (*CHI '21*). Association for Computing Machinery, New York, NY, USA, Article 263, 15 pages. <https://doi.org/10.1145/3411764.3445533>
- [16] Jong-Hyun Kim and Jung Lee. 2020. Controlling your contents with the breath: Interactive breath interface for VR, games, and animations. *Plos one* 15, 10 (2020), e0241498.
- [17] Takahiro Kusabuka and Takuya Indo. 2020. IBUKI: Gesture Input Method Based on Breathing. In *Adjunct Publication of the 33rd Annual ACM Symposium on User Interface Software and Technology*, 102–104.
- [18] James R Lewis. 2006. Usability testing. *Handbook of human factors and ergonomics* 12 (2006), e30.
- [19] Ryoma Ogawa, Kyosuke Futami, and Kazuya Murao. 2021. NasalBreathInput: A Hands-Free Input Method by Nasal Breath Gestures using a Glasses Type Device. In *The 23rd International Conference on Information Integration and Web Intelligence*, 620–624.
- [20] Ryoya Onishi, Tao Morisaki, Shun Suzuki, Saya Mizutani, Takaaki Kamigaki, Masahiro Fujiwara, Yasutoshi Makino, and Hiroyuki Shinoda. 2021. DualBreath: Input Method Using Nasal and Mouth Breathing. In *Augmented Humans Conference 2021*, 283–285.
- [21] Pablo E Paredes, Yijun Zhou, Nur Al-Huda Hamdan, Stephanie Balters, Elizabeth Murnane, Wendy Ju, and James A Landay. 2018. Just breathe: In-car interventions for guided slow breathing. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 2, 1 (2018), 1–23.
- [22] Shwetak N Patel and Gregory D Abowd. 2007. Blui: low-cost localized blowable user interfaces. In *Proceedings of the 20th annual ACM symposium on User interface software and technology*, 217–220.
- [23] Rakesh Patibanda, Florian'Floyd' Mueller, Matevz Leskovsek, and Jonathan Duckworth. 2017. Life tree: understanding the design of breathing exercise games. In *Proceedings of the annual symposium on computer-human interaction in play*, 19–31.
- [24] Michael Brent Phillips, Jason Beach, Michael Cathey, Jake Lockert, and William Satterfield. 2017. Reliability and validity of the Hexoskin wearable body metrics telemetry shirt. *Journal of Sport and Human Performance* 5, 2 (2017).
- [25] Mirjana Prpa, Ekaterina R. Stepanova, Thecla Schiphorst, Bernhard E. Riecke, and Philippe Pasquier. 2020. Inhaling and Exhaling: How Technologies Can Perceptually Extend Our Breath Awareness. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (*CHI '20*). Association for Computing Machinery, New York, NY, USA, 1–15. <https://doi.org/10.1145/3313831.3376183>
- [26] Gabriel Reyes, Dingtian Zhang, Sarthak Ghosh, Pratik Shah, Jason Wu, Aman Parnami, Bailey Bercik, Thad Starner, Gregory D Abowd, and W Keith Edwards. 2016. Whoosh: non-voice acoustics for low-cost, hands-free, and rapid input on smartwatches. In *Proceedings of the 2016 ACM International Symposium on Wearable Computers*, 120–127.
- [27] Daniela Roditi and Michael E Robinson. 2011. The role of psychological interventions in the management of patients with chronic pain. *Psychology research and behavior management* 4 (2011), 41.
- [28] Jeff Sauro. 2012. things to know about the Single Ease Question (SEQ). *Measuring U* 2012 (2012).
- [29] Misha Sra, Xuhai Xu, and Pattie Maes. 2018. Breathvr: Leveraging breathing as a directly controlled interface for virtual reality games. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*, 1–12.
- [30] Markus Tatzgern, Michael Domhardt, Martin Wolf, Michael Cenger, Gerlinde Emshuber, Radomir Dinic, Nathalie Gerner, and Arnulf Hartl. 2022. AirRes Mask: A Precise and Robust Virtual Reality Breathing Interface Utilizing Breathing Resistance as Output Modality. In *CHI Conference on Human Factors in Computing Systems*, 1–14.
- [31] Paul Tennent, Duncan Rowland, Joe Marshall, Stefan Rennick Egglestone, Alexander Harrison, Zachary Jaime, Brendan Walker, and Steve Benford. 2011. Breathalising games: understanding the potential of breath control in game interfaces. In *Proceedings of the 8th international conference on advances in computer entertainment technology*, 1–8.
- [32] Robby van Delden, Danny Plass-Oude Bos, Antje Jacoba Vivian de With, Koen Vogel, Randy Klaassen, Nynke Zwart, Joyce Faber, Boony Thio, and Mattienne van der Kamp. 2020. SpiroPlay, a Suite of Breathing Games for Spirometry by Kids & Experts. In *Proceedings of the Annual Symposium on Computer-Human Interaction in Play*, 400–413.
- [33] Vincent Van Rheden, Eric Harbour, Thomas Finkenzeller, Lisa Anne Burr, Alexander Meschtscherjakov, and Manfred Tscheligi. 2021. Run, Beep, Breathe: Exploring the Effects on Adherence and User Experience of 5 Breathing Instruction Sounds while Running. In *Proceedings of the 16th International Audio Mostly Conference, AM 2021 (ACM International Conference Proceeding Series)*. Association for Computing Machinery, 16–23. <https://doi.org/10.1145/3478384.3478412> Publisher Copyright: © 2021 ACM.; 16th International Audio Mostly Conference, AM 2021 ; Conference date: 01-09-2021 Through 03-09-2021.
- [34] X Rosalind Wang, Joseph T Lizier, Amalia Z Berna, Florence G Bravo, and Stephen C Trowell. 2015. Human breath-print identification by E-nose, using information-theoretic feature selection prior to classification. *Sensors and Actuators B: Chemical* 217 (2015), 165–174.
- [35] Motoji Yamamoto, Takeshi Ikeda, and Yoshinobu Sasaki. 2008. Real-time analog input device using breath pressure for the operation of powered wheelchair. In *2008 IEEE International Conference on Robotics and Automation*. IEEE, 3914–3919.
- [36] Andrea Zaccaro, Andrea Piarulli, Marco Laurino, Erika Garbella, Danilo Menicucci, Bruno Neri, and Angelo Gemignani. 2018. How Breath-Control Can Change Your Life: A Systematic Review on Psycho-Physiological Correlates of Slow Breathing. *Frontiers in human neuroscience* 12 (2018), 353. <https://doi.org/10.3389/fnhum.2018.00353>